CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Tech. Examination (EE)

April-2013

EE-204/EE-204.01 ELCTRICAL POWER SYSTEM-I

Date: 24.04.2013, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) What do you mean by grading of cables? Explain Intersheath grading with its disadvantages. **[07]**

Q - 2 (a) Give the classification of cables. Explain Belted cable in detail with neat sketch. **[07]**

(b) Which are the desirable properties of good conductor? Give the comparison between Copper and Aluminum conductor. **[04]**

(c) Discuss advantages and disadvantages of DC transmission system over an AC transmission system. **[03]**

OR

Q - 2 (a) In a 3-phase overhead system, each line is suspended by a string of 3 insulators. The voltage across the top unit(i.e. near the tower) and middle unit are 10kV and 11kV Respectively. Calculate (i) the ratio of shunt capacitance to self capacitance of each Insulator, (ii) the string efficiency and (iii) line voltage. **[07]**

(b) Derive the equation of the capacitance of a single core cable. **[05]**

(c) What are the advantages and disadvantages of Oil filled cable? **[02]**

Q - 3 (a) Explain synchronous condenser method of power factor improvement with its advantages and disadvantages. **[06]**

(b) A consumer has an average demand of 400 kW at a p.f. of 0.8 lagging and annual load factor of 50%. The tariff is Rs 50 per kVA of maximum demand per annum plus 5 paise per kWh. If the power factor is improved to 0·95 lagging by installing phase **[06]**

advancing equipment, calculate : (i) the capacity of the phase advancing equipment ii) The annual saving effected.

The phase advancing equipment costs Rs 100 per kVAR and the annual interest and depreciation together amount to 10%.

(c) Explain the term : (i) Demand factor (ii) Diversity factor [02]

OR

Q - 3 (a) What is tariff? Explain Flat rate, Power factor and Two-part tariff in detail. [07]

(b) The equipment in a power station costs Rs 15, 60,000 and has a salvage value of Rs 60,000 at the end of 25 years. Determine the depreciated value of the equipment at the end of 20 years on the following methods : [07]

(i) Straight line method ; (ii) Diminishing value method ; (iii) Sinking fund method at 5% compound interest annually.

SECTION – II

Q - 4 (a) Explain Nominal T method of transmission line with its Phasor diagram. [07]

Q - 5 (a) Derive the equation of Inductance of a 3-phase overhead line with unsymmetrical spacing. [07]

(b) A 3-phase, 50Hz, 150 km line has a resistance, inductive reactance and capacitive shunt admittance of 0.1 Ω 0.5 Ω and $3*10^{-6}$ S per km per phase. If the line delivers 50 MW at 110 kV and 0.8 p.f. lagging, determine the sending end voltage and current. **Assume nominal π circuit for the line.** [07]

OR

Q - 5 (a) Using Rigorous method, derive expression for sending end voltage and current for a long transmission line. [07]

(b) A 3-phase overhead transmission line has a series impedance of (10 + j30) Ω per phase. For receiving end and sending end voltages of 132 kV and 140kV respectively. Draw receiving end power circle diagram. Determine (i) The maximum real power delivered by the line and the load power factor under that condition. (ii) The capacity of shunt compensation equipment for supplying a load of 150 MVA at 0.8 p.f. lagging and the power angle (torque angle) under that condition. [07]

Q - 6 (a) Find the following for a single circuit transmission line delivering a load of 50 MVA at 110 kV and p.f. 0·8 lagging : (i) sending end voltage (ii) sending end current (iii) sending end power (iv) transmission efficiency. [07]

Given A = D = 0·98 ∠3°; B = 110 ∠75° ohm; C = 0·0005 ∠80° siemen.

(b) Write a short note on Ferranti effect. [05]

(c) Write equation of capacitance of 1-phase transmission line considering the effect of earth. [02]

<u>OR</u>

Q - 6 (a) A 3-phase, 50 Hz, 110 kV, transmission line has flat horizontal spacing with 4 m between adjacent conductors. The conductor diameter is 1.05 cm. Find the capacitance to neutral and the charging current per km of line. Assume full transposition. [07]

(b) Derive the equation of capacitance to ground for 1-phase Two-wire line. [05]

(c) Explain the importance of transposition of transmission lines. [02]

****